Atlanta Botanical Garden - Expansion

Northern Red Oak

Quercus rubra

Family: Not stated Also known as: —

GARDENS Passage: 1 **TOTAL 1**



Summer / form
Library candidate



Fall color
Library candidate

Not available

Flower
Not available



Leaf close-up
Library candidate

QUICK FACTS & PLANTING

Type: Tree
Evergreen/Deciduous: Deciduous
Mature size: 60–90 ft (up to 100+ ft). Confirmed by USDA (mature ~81 ft), Wildflower.org (75–100 ft), multiple sources (50–75 to 65–92 ft ranges). × 50–75 ft. Confirmed.
Hardiness / Zone 8a: Suitable for Zone 8a. Hardy Zones 3–8. Native to eastern North America including Georgia. Performs in Atlanta with deep, acidic, well-drained soil. Fast-growing for an oak. Transplants more readily than many oaks. Good shade tree. Fall color russet-red reliable in Zone 8a when soil is suitable. Prefers protection from extreme afternoon heat when young in hotter microclimates. Restart pass (2026-10-13): NC State confirms 50–75 ft, Piedmont and mountain preference, mesic slopes. [https://plants.ces.ncsu.edu/plants/quercus-rubra/]
Sun: Full sun to part shade. One of the more shade-tolerant oaks.
Water: Medium. Prefers moist, well-drained soil. Moderately drought tolerant once established. Low drought tolerance when young in some sources.
Soil: Deep, well-drained, acidic preferred (pH 4.3–7.0). Adaptable to loam, clay, sand. Avoid alkaline or shallow rocky soils (chlorosis risk).
Growth rate: Not stated
Not stated
Planting:
Siting: Plant in full sun to part shade in deep, moist, acidic, well-drained soil. Water regularly until established (first 3–5 years). Transplants more easily than many oaks due to limited taproot. Requires ample space. Suitable as large shade, park, or street tree where space allows. Minimal pruning. Long-lived timber and wildlife tree.
Soil prep: Deep, well-drained, acidic preferred (pH 4.3–7.0). Adaptable to loam, clay, sand. Avoid alkaline or shallow rocky soils (chlorosis risk).
Depth/spacing: Not stated
Season: Not stated
Establishment: Not stated

ORIGIN, CHARACTERISTICS & HEALTH

Origin / native: Mesic upland forests, ravines, north and east slopes, well-drained loamy sands. Prefers deep soils.
Form / habit: Deciduous. Large tree. Pyramidal when young, becoming broad rounded with age. Moderate to fast growth for an oak. Long-lived.
Foliage: Russet to red, yellow-brown. Reliable in Zone 8a with good conditions.
Flowers / fruit: Not stated
Bark / stem: Dark gray-brown with flat-topped ridges separated by shallow furrows. Older bark ridged.
Not stated
Health:
Health notes: Possible oak wilt (susceptible; avoid pruning in infested areas during growing season), anthracnose, leaf blister, powdery mildew, Nectria canker, Armillaria. Generally durable and pollution-tolerant. Chlorosis in high pH. **Open:** Specific Zone 8a incidence of oak wilt and other diseases is thin.
Not stated
Additional:
- Pointed, bristle-tipped lobes (red oak group). - Acorns important for wildlife. - Fast-growing relative to other oaks. - Withstands pollution. - Valuable timber tree. - One of the most important commercial oaks.